

Pattern making using looping

Left angle triangle and pyramid

```
print('numbers- rows- Left angle triangle')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(i, end=' ')
    print()
```

```
numbers- rows- Left angle triangle
-----
      1
     2 2
    3 3 3
   4 4 4 4
  5 5 5 5 5
```

```
print('number- column- left angle triangle')
print('-'*36)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(j, end=' ')
    print()
```

```
number- column- left angle triangle
-----
      0
     0 1
    0 1 2
   0 1 2 3
  0 1 2 3 4
```

```
print('star- left angle triangle')
print('-'*30)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print('*', end=' ')
    print()
```

```
star- left angle triangle
-----
      *
     * *
    * * *
   * * * *
  * * * * *
```

```
print('upper case- row- left angle triangle')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64), end=' ')
    print()
```

```
upper case- row- left angle triangle
-----
      A
     B B
    C C C
   D D D D
  E E E E E
```

```
print('upper case- column- left angle triangle')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65), end=' ')
    print()
```

upper case- column- left angle triangle

```
-----
A
A B
A B C
A B C D
A B C D E
```

```
] print('lower case- row- left angle triangle')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64+32), end=' ')
    print()
```

lower case- row- left angle triangle

```
-----
a
b b
c c c
d d d d
e e e e e
```

```
] print('lower case- column- left angle triangle')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65+32), end=' ')
    print()
```

lower case- column- left angle triangle

```
-----
a
a b
a b c
a b c d
a b c d e
```

```
print('numbers- rows- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(i, end=' ')
    print()
```

numbers- rows- pyramid

```
-----
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
print('numbers- column- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(j, end=' ')
    print()
```

numbers- column- pyramid

```
-----
0
0 1
0 1 2
0 1 2 3
0 1 2 3 4
```

```
print('star- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print('*', end=' ')
    print()
```

star- pyramid

```
-----
*
* *
* * *
* * * *
* * * * *
```

```
print('upper-row- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64), end=' ')
    print()
```

star- pyramid

```
-----
A
B B
C C C
D D D D
E E E E E
```

```
print('upper-column- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65), end=' ')
    print()
```

upper-column- pyramid

```
-----
A
A B
A B C
A B C D
A B C D E
```

```
print('lower-column- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65+32), end=' ')
    print()
```

lower-column- pyramid

```
-----
a
a b
a b c
a b c d
a b c d e
```

```
print('lower-row- pyramid')
print('-'*35)
for i in range(1,6):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64+32), end=' ')
    print()
```

lower-row- pyramid

```
-----
a
b b
c c c
d d d d
e e e e e
```

```
[122]: for i in range(6,0,-1):
        for k in range(6,i,-1):
            print(' ', end='')
        for j in range(0,i):
            print(chr(i+64+32), end='')
        print()
```

```
ffffff
eeeee
dddd
ccc
bb
a
```

```

print('inverse numbers- rows- Left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(i, end=' ')
    print()

```

inverse numbers- rows- Left angle triangle

```

-----
6 6 6 6 6
5 5 5 5
4 4 4 4
3 3 3
2 2
1

```

```

print('inverse numbers- column- Left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(j, end=' ')
    print()

```

inverse numbers- column- Left angle triangle

```

-----
0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
0 1
0

```

```

print('inverse upper case- row- left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64), end=' ')
    print()

```

inverse upper case- row- left angle triangle

```

-----
F F F F F
E E E E
D D D D
C C C
B B
A

```

```

print('inverse upper case- column- left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65), end=' ')
    print()

```

inverse upper case- column- left angle triangle

```

-----
A B C D E F
A B C D E
A B C D
A B C
A B
A

```

```

print('Inverse lower case- row- left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(i+64+32), end=' ')
    print()

```

Inverse lower case- row- left angle triangle

```

-----
f f f f f
e e e e e
d d d d
c c c
b b
a

```

```

print('Inverse lower case- column- left angle triangle')
print('-'*35)
for i in range(6,0,-1):
    for k in range(6,i,-1):
        print(' ', end=' ')
    for j in range(0,i):
        print(chr(j+65+32), end=' ')
    print()

```

Inverse lower case- column- left angle triangle

```

-----
a b c d e f
a b c d e
a b c d
a b c
a b
a

```